SQL Exercises - Students Table Answers

CREATE TABLE

```sql
CREATE TABLE Students (
    StudentID INT PRIMARY KEY,
    FirstName NVARCHAR(50),
    LastName NVARCHAR(50),
    BirthDate DATE,
    GradeLevel INT
);
```

INSERT DATA

```sql
INSERT INTO Students (StudentID, FirstName, LastName, BirthDate, GradeLevel)
VALUES
(3, 'Michael', 'Johnson', '2007-03-15', 7),
(4, 'Emily', 'Davis', '2005-07-20', 9),
(5, 'Matthew', 'Williams', '2006-11-12', 8),
(6, 'Olivia', 'Brown', '2008-02-25', 6),
(7, 'Daniel', 'Garcia', '2007-05-30', 7),
(8, 'Sophia', 'Martinez', '2005-12-14', 9),
(9, 'Liam', 'Lopez', '2006-01-08', 8),
(10, 'Emma', 'Wilson', '2008-06-05', 6);
```

Exercises

1. Retrieve All Records: Retrieve every column and row from the Students table.
2. Retrieve Specific Columns: Select only the FirstName, LastName, and GradeLevel columns from the Students table.
3. Filter by Grade Level: Retrieve records for students who are in GradeLevel 7.
4. Filter by Birth Date: Retrieve records for students born after '2007-01-01'.
5. Filter by Last Name: Retrieve records for students whose LastName is 'Brown'.
6. Order by First Name: Retrieve all student records and order them alphabetically by FirstName in ascending order.
7. Order by Birth Date: Retrieve all student records and sort them by BirthDate from oldest to youngest.
8. Limit the Number of Rows Returned: Retrieve the 3 youngest students by ordering the table by BirthDate in descending order and limiting the result to 3 rows.
9. Count Total Number of Students: Count and return the total number of records in the Students table.
10. Calculate the Average Grade Level: Compute and return the average GradeLevel of all students.
11. Find the Minimum Birth Date: Retrieve the earliest BirthDate in the table.
12. Find the Maximum Grade Level: Retrieve the highest GradeLevel in the table.
13. Group Students by Grade Level: Group the records by GradeLevel and count how many students are in each grade.
14. Update a Student's Grade Level: Update the GradeLevel of the student whose StudentID is 6.
15. Update a Student's Last Name: Change the LastName of the student whose StudentID is 8.
16. Delete a Student Record: Delete the record of the student whose StudentID is 10.
17. Select Students Within a Birth Date Range: Retrieve records for students whose BirthDate is between '2006-01-01' and '2007-12-31'.
18. Retrieve Students Whose First Name Starts with E: Retrieve all records where FirstName starts with the letter 'E'.

Answer Key

1. 1. Retrieve All Records

```
SELECT * FROM Students;
```

2. 2. Retrieve Specific Columns

```
SELECT FirstName, LastName, GradeLevel
FROM Students;
```

3. 3. Filter by Grade Level

```
SELECT *
FROM Students
WHERE GradeLevel = 7;
```

4. 4. Filter by Birth Date

```
SELECT *
FROM Students
WHERE BirthDate > '2007-01-01';
```

5. 5. Filter by Last Name

```
SELECT *
FROM Students
WHERE LastName = 'Brown';
```

6. 6. Order by First Name

```
SELECT *
FROM Students
ORDER BY FirstName ASC;
```

7. 7. Order by Birth Date

```
SELECT *
FROM Students
ORDER BY BirthDate ASC;
```

8. 8. Limit the Number of Rows Returned

```
SELECT TOP 3 *
FROM Students
ORDER BY BirthDate DESC;
```

9. 9. Count Total Number of Students

```
SELECT COUNT(*) AS TotalStudents
FROM Students;
```

10. 10. Calculate the Average Grade Level

```
SELECT AVG(GradeLevel) AS AverageGradeLevel
FROM Students;
```

11. 11. Find the Minimum Birth Date

```
SELECT MIN(BirthDate) AS EarliestBirthDate
FROM Students;
```

12. 12. Find the Maximum Grade Level

```
SELECT MAX(GradeLevel) AS HighestGradeLevel
FROM Students;
```

13. 13. Group Students by Grade Level

```sql
SELECT GradeLevel, COUNT(*) AS StudentCount
FROM Students
GROUP BY GradeLevel;
```

14. 14. Update a Student's Grade Level

```sql
UPDATE Students
SET GradeLevel = 7
WHERE StudentID = 6;
```

15. 15. Update a Student's Last Name

```sql
UPDATE Students
SET LastName = 'Anderson'
WHERE StudentID = 8;
```

16. 16. Delete a Student Record

```sql
DELETE FROM Students
WHERE StudentID = 10;
```

17. 17. Select Students Within a Birth Date Range

```sql
SELECT *
FROM Students
WHERE BirthDate BETWEEN '2006-01-01' AND '2007-12-31';
```

18. 18. Retrieve Students Whose First Name Starts with E

```sql
SELECT *
FROM Students
WHERE FirstName LIKE 'E%';
```